

Pregnant in Prison— The Incarcerated Woman's Experience: A Preliminary Descriptive Study

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This is an exploratory study of depression in pregnant women incarcerated in a state prison in California. It was hypothesized that the inmates would experience higher levels of postpartum depression than would nonincarcerated women who have recently given birth. Participants were 120 inmates who recently gave birth while incarcerated. They were interviewed and given the Beck Depression Inventory. It was found that none of the participants were clinically depressed. Descriptive statistics are presented in detail. A qualitative component of the study explored the life experiences of pregnancy by participants. Future research will empirically address the lack of depression or negative reaction to incarcerated pregnancy.

Keywords: female inmates; pregnant inmates; postpartum depression; correctional health

Female offenders are the most rapidly growing segment of the ever-increasing prison population, yet research on incarcerated women is limited (Church, 1990; Egley, Miller, Granados, & Ingram-Fogel, 1992; Fogel, 1993; LeFlore & Holston, 1990). In particular, little is known about women prisoners' experiences with motherhood and pregnancy during incarceration.

Incarceration creates a unique and complex set of problems for these women and their children and families. A 1990 study conducted by Church found that as many as 25% of women entering prison were either pregnant or had recently given birth. In 1987, the average age of incarcerated mothers in the Mississippi Department of Corrections was 29 years, and the average number of children per mother was three (LeFlore & Holston, 1990). These children may or may not be in the mother's custody. These women typically lack social support from friends and family. They are often single, poor women from minority backgrounds ill prepared for child rearing. In spite of this, they are usually the primary caregivers of their children before entering prison (LeFlore & Holston, 1990). These facts raise many questions: How do incarcerated women feel about being pregnant in prison? How do they care for their children? With whom do their children live? Do they have regular contact with their children? Will they be with their children when they are released from prison?

Valley State Prison for Women (VSPW) in Chowchilla, California (rural central California), houses the largest pregnant incarcerated population in the state. Between January

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1998 and December 2001, VSPW housed an average of 100 pregnant inmates per month, and an average of 16 babies per month were delivered. Women may be incarcerated at any point during their pregnancy. While incarcerated, inmates often have limited resources both inside and outside of prison to assist them in making decisions about who will care for their unborn child. Within the California Department of Corrections (CDC), inmates are given 2 to 3 days with their newborn. Before birth, social services will try to arrange care for the child, usually with a blood relative, the father of the child, or a notarized legal guardian. In other cases, the infant is placed in the care of Child Protective Services.

There are few studies about the incarcerated pregnant woman's experience, and studies of their postpartum phase of motherhood are nonexistent (Egley et al., 1992; Fogel, 1993; LeFlore & Holston, 1990). Considering this population's environment and situation, it would seem that pregnancy would increase the stress levels and negative experience, resulting in a poor experience of pregnancy and possibly depression during or after the pregnancy. Is this a population in which higher levels of postpartum depression would be found? It seems as though this could be a possibility considering all of the above factors. Therefore, this study explored depression and postpartum depression among incarcerated women after giving birth. A discussion of research related to postpartum depression will highlight these factors.

Literature Review

The frequency rates of postpartum depression in nonincarcerated populations range from 3% to 20%, with a 10% to 35% rate of recurrence (Kennedy & Suttentfield, 2001; Kruckman & Smith, 1998). Transitory blues occurs in 50% to 80% of births (Kruckman & Smith, 1998) and typically lasts fewer than 2 weeks (Kennedy & Suttentfield, 2001). Zerkowicz and Milet (1995) report the prevalence of postpartum depression in the general community in Quebec, Canada, to be 6.2%. Women who did not work outside the home and those with blue collar or unskilled jobs had higher rates of depression compared to women with higher status jobs. Personality variables were also shown to play a role in postpartum depression. Lack of role clarity, feelings of inadequacy regarding child bearing, and ambivalence about the pregnancy have been shown to be predictive of emotional problems, such as depression (Nilsson & Almgren, 1970; Tentoni & High, 1980). Low self-esteem, age, and environmental conditions, such as poor socioeconomic conditions and lack of financial resources, were shown to have an impact on postpartum depression (Uddenbrug & Nilsson, 1975).

Lee (1997) states that postpartum depression is generally milder than other depressive disorders. She reported the prevalence rates cross-culturally to be 6% to 10%. Lee believes that this low rate is support for the argument that postpartum depression may be on a continuum of adjustment to motherhood and major life change. "Major life-events, particularly those involving the partner or close family, have been shown to increase the risk of postpartum depression in several cultures" (Lee, 1997, p. 97). Lee also discusses the experience of fathers and how they commonly experience the same configuration of depressive symptoms, counteracting the belief that postpartum depression is caused by major shifts in hormonal levels.

Chalmers and Chalmers (1986) report that women with postpartum depression have similar concerns about baby care, the motherhood role, the woman's self-concept, and difficulties in the marital relationship. For the women with postpartum depression, these concerns are intensified to a level that they can "be associated with emotional, intellectual, or behavioral disturbances in their adjustment to parenthood" (Chalmers & Chalmers, 1986, p. 97). It would seem that pregnant incarcerated women would have some of the same concerns because of the stressors of being incarcerated while their baby is being taken care of by others. At the same time, the stigma of being a prisoner does not fit well with the

concept of motherhood in the eyes of the community, possibly increasing the inmate's stress on reentering the community. This population has to integrate *felon* with *woman* and *mother* while possibly being depressed or suffering emotional, intellectual, or behavioral problems.

LeFlore & Holston (1990) found that "incarcerated mothers are as acculturated into the parenting role as are noncriminal mothers" (p. 18) and are expected to continue their role as mother in the household upon their release. This study contributes the fact that incarcerated mothers are aware of what is expected of them without addressing the many complicated issues of incarceration's impact on the mothers' state of mind, parenting abilities, or coping skills.

Stanton (1980) said an estimated one third of incarcerated mothers never write to their children, and 55% had no visits with their children, highlighting the level of contact with existing children.

Fogel (1993) conducted an exploratory descriptive study of incarcerated pregnant women to document risk factors and pregnancy outcomes. It was found that depression levels were higher in pregnant inmates compared to those reported for all female inmates and incarcerated mothers. Yet Fogel also found "a low incidence of poor pregnancy outcomes," which was described as "surprising in light of the high levels of pregnancy risk factors reported" (p. 38). Furthermore, Egley et al. (1992) found that inmates are less likely to deliver prematurely than are nonincarcerated control groups. These findings make a positive statement for the medical processes for the pregnant inmates, who may not have had the same level of access to medical care if not incarcerated. These studies describe difficult emotional experiences for pregnant inmates but more positive birthing outcomes in terms of health of the baby and normal completion of the pregnancy.

In another study, Hayworth et al. (1980) found that high anxiety and high hostility are related to postpartum depression. The younger participants in their sample and those who perceived they have less control of their lives were more likely to be depressed. The most depressed women were more hostile before giving birth and extrapunitive, meaning they direct their hostility toward others. Hostility and anxiety are both commonly expressed by incarcerated women, leading to an expectation that postpartum depression levels would be greater in this population.

These studies begin to describe pregnancy for incarcerated women. The present study further explores and describes this experience for female prisoners.

Research Questions

1. Will incarcerated female prisoners who have recently given birth while incarcerated have a rate of postpartum depression higher than the average rate of postpartum depression in the general, nonincarcerated community?
2. Is there a relationship between postpartum depression and (a) the number of times the mother is incarcerated; (b) the mother's age at first pregnancy; (c) the number of deceased children; (d) the number of miscarriages; (e) the number of children in the mother's custody; (f) the location of the child, e.g., living with father, grandmother, or in foster care; (g) the length of time with the newborn before mother and baby are separated; (h) the number of times the mother has seen the baby since the birth; and (i) the amount of contact the mother will have with the baby after release from prison.

We also describe four basic areas: (1) life experiences of pregnancy, (2) postbirth contact with newborns and future logistics, (3) future plans for pregnancy, and (4) thoughts about their most recent pregnancy while incarcerated.

Method

Participants

Participants in this study were 120 inmates at VSPW who had recently given birth while incarcerated. Valley State Prison for Women is one of the two largest women's prisons in the world, and holds the world's largest pregnant incarcerated population. All pregnant prisoners from northern California are housed at VSPW, where the average number of pregnant inmates is 1,240.5 in a year and 103.4 in a month. The overall prison population averages 3,000 to 3,800 inmates, and was 3,200 across the time frame of this study. The time period of the data collection for this study was 1999 to 2002.

The average age of the sample was 29.1 years, with a range of 19 to 41 years. The average sentence length was 2 years 3 months, ranging from 0.1 months left of the sentence to 12.5 years. The average number of incarcerations was reported to be 4.1. The general racial makeup of the prison was 30% White, 30% African American, 35% Hispanic, and 5% other groups (CDC, 1999). Further description of the sample is presented in the study results. It is important to note that the study participants were all highly functioning inmates who were not seriously mentally ill, developmentally disabled, or otherwise disabled.

Procedure

This study used archival data that was originally collected to screen female inmates for possible participation in a postpartum support group. Inmates who were identified by the obstetric clinic as having recently given birth were interviewed. A structured interview format was used. The Beck Depression Inventory was given to the inmates to read and fill out on their own. When this inventory was completed, the experimenter invited the inmate to join the postpartum support group. This procedure typically took 25 minutes to complete.

Instruments

Postpartum interview form. This is a structured interview developed by the lead author to screen for postpartum depression among women in the prison. The questionnaire explores the experience of pregnancy in the prison, with inquiry into the following areas: life experiences of pregnancy, postbirth contact with newborns and future logistics, future plans for pregnancy, and thoughts about their most recent pregnancy while incarcerated.

Beck Depression Inventory. This instrument is well known and widely used to screen for depression and to measure depression levels in individual persons. It is accepted in the psychological community as a strong, reliable, and valid measure. In the VSPW study, it was not intended to be the final instrument to diagnose depression. Previous screening of participants had ruled out serious mental illness, including major depressive disorder. Rather, the Beck Depression Inventory was used only as a tool to screen for the group and to give subjective data of the possible levels of depression, and the results were interpreted in light of more objective clinical data.

Results

Descriptive Results

Across 4 years, from 1999 to 2002, the average number of pregnant inmates at the prison each year was 1240.5, with a mean of 103.4 pregnant inmates incarcerated each month. In the same

Table 1. Descriptive Statistics of Incarcerated Women Who Recently Gave Birth

	<i>M</i>	<i>SD</i>	Range
Age, years	29.1	5.0	19-41
Length of sentence, years	2.3	1.8	0.1-12.5
Times incarcerated	4.1	6.3	1-50
Age of first pregnancy, years	18.5	3.4	14-31
Number of pregnancies	5.3	2.9	1-15
Number of living children	3.6	2.1	0-10
Number of deceased children	0.5	1.3	0-11
Number of miscarriages	0.8	1.2	0-7
Number of abortions	0.9	1.2	0-5
Number of children in mother's custody	2.1	1.9	0-9
Number of weeks pregnant	35.8	8.9	5-45
Time with child after birth, days	3.4	6.3	0-60
Visits since birth	0.3	0.9	0-6
Usage of birth control	1.8	1.1	0-6
Marital status	Percentage		
Single	73		
Married (includes common law relationships)	19		
Separated or divorced	9		
Widowed	0		

time period, the average number of deliveries per year was 197, with 16.3 mean deliveries per month. For comparison, the average inmate population was approximately 3,200 inmates.

The mean Beck Depression Inventory score was 17.3. It was found that 40% of study participants scored in the minimal range; 20%, in the mild range; 23%, in the moderate range; and 17%, in the severe range. Therefore, 40% scored in the moderate to severe range, and 60% scored in the mild or minimal range.

General descriptive information is shown in Table 1. Items meriting comment are noted here. First, the accuracy of data concerning self-reported number of times incarcerated (4.1 times) is highly questionable because some inmates may have included incarcerations in county facilities before coming to prison. Second, the average pregnancy completed at the prison was full-term, and only 9% of the babies needed special medical attention after birth. Third, the mean number of mother-baby visits after the mother returned to prison was 0.3. Many inmates reported that they did not want their newborn brought to the prison because they were too young and vulnerable to disease.

With regard to marital status for this sample, it was reported that 73% were single, 18% were married, 2% were separated, 7% were divorced, and 1% reported common law relationships. The inmates were asked about their plans to have future children; 51% said no, 31% said yes, and 19% were undecided. The participants also were asked about prevention of future pregnancy if they chose not to have more children. It was reported that they used 1.8 forms of birth control, when asked to choose from 11 possible choices. Sixty-two percent of the inmates planned to use birth control, 24% wanted a Tubule Ligation, ie tubes tied, (many asked the prison to do this after they gave birth, but this was usually denied because of the cost of the procedure), 6% had another plan for prevention, and 8% had no plans for prevention.

The participants were asked about the location of the baby after giving birth. At the hospital, a family member or a social worker is required to be available to pick up the baby, and the inmate is returned to prison after release from the hospital. Babies were taken to various

Table 2. Present Location of the Child (%)

With the father	17
With immediate family	41
With extended family	12
Foster care, but inmate has custody	9
Child Protection Services permanent	6
Adopted	2
Other	6
Abortion	6
Miscarriage	2

Note: *N* = 113.

Table 3. Means of Financial Support (%)

Find a job	44
School or training and then find a job	5
Supported by husband or boyfriend who works	17
Supported by family member	3
AFDC, county support, WIC	18
Entering drug treatment program	2
Savings account	1
Other	4
Don't know	7

Note: *N* = 106. AFDC = Aid to Families With Dependent Children; WIC = Women, Infants, and Children.

locations, as shown in Table 2. Six percent indicated locations other than the choices given; this group includes possible fetal deaths, stillborn births, or neonatal death.

When asked about their level of interaction with the baby upon their release from prison, 78% of participants reported that they would have full custody, 3% had visitation rights only, 2% had only access to information about their baby without visitation rights, 4% planned to attempt reunification, 1% planned to go to the Mother-Infant Program, and 13% said the status was unknown at the time of the interview.

The fathers' role in the babies' lives also was measured: 32% of the fathers had a full-time role, 15% had a part-time role, 41% played no part (this includes fathers who did not know of the pregnancy), and 13% were incarcerated at the time of the interview.

To support themselves after release from prison, nearly half of the study participants planned to get a job, either right away (44%) or after attending school or training (5%). Other anticipated means of financial support are reported in Table 3. When asked about their means of emotional support, participants cited many forms and combinations of forms of support, as shown in Table 4.

Seven of 10 study participants reported drug use; 30% reported no drug use, and 2% reported none while pregnant. The most widely used drug of choice was methamphetamine or crank (21%), followed by cocaine (13%), alcohol (10%), marijuana (8%), and tobacco (1%). Sixteen percent reported use of more than one drug type. No participants cited heroin or PCP as a drug of choice. It was determined that 60% of the inmates had been educated on the effects of drugs on pregnancy, through drug rehabilitation programs or high school classes.

Table 4. Means of Emotional Support (%)

Three or more sources of support	7
Family and friends	12
Family and husband or boyfriend	7
Friends only	3
Husband or boyfriend only	8
Family only	43
No one	13
Other	7

Note: $N = 98$.

Correlational Results

It was hypothesized that a relationship would be found between depression and several variables. Results indicate that age is related to the number of pregnancies ($r = .63$), and that number of pregnancies is related to (a) number of living children ($r = .73$), (b) number of miscarriages ($r = .57$), and (c) number of abortions ($r = .52$). Number of living children is related to the number of children in custody ($r = .50$), and number of deceased children is related to miscarriages ($r = .50$). None of these findings are surprising, but they are reported to show consistency.

Depression was found to be affected by several factors. A significant difference was found between depression and (a) number of pregnancies ($r = .35$), (b) length of sentence ($r = .16$), (c) number of miscarriages ($r = .40$), and (d) number of children in custody ($r = .31$). However, these are weak differences. Also, a significant weak difference was found between number of children in the mother's custody and her age at first pregnancy ($r = .02$). A strong significant difference was found between depression and number of deceased children ($r = .66$), as well as drug of choice ($r = .67$).

Qualitative Results

Qualitative results were obtained through categorization of responses to the open-ended questions as either positive or negative. The questions addressed the inmate's experience with the medical process (7 positive reports, 1 negative report), feelings about the prison environment (5 negative reports), interactions between prison staff and inmates (3 negative statements), separation from their baby (15 negative statements, 0 positive statements), and their total experience of being pregnant while incarcerated (6 positive reports, 20 negative reports). Only 57 inmates in the sample expressed concerns of these kinds, with 22.8% expressing a positive experience and 77.2% expressing a negative experience.

The following statements summarize comments repeated by several inmates during the open-ended question section of the interview:

1. It is stressful to be pregnant in prison.
2. It is difficult to be in the reception center (the area where they stay for their first 3 months in prison) because they are allowed no phone calls and are locked down most of the day.
3. Of those who mentioned medical care, most were pleased with it and said it was good care and a positive part of their experience.
4. All agreed that the environment was a poor place to be pregnant.
5. Although a small number of inmates said that their incarceration while pregnant was a positive experience, most felt that it was not.

6. Most expressed a painful emotional experience due to separation from their baby.
7. Almost all said that what they would change is to not come to prison.

Discussion

The purpose of this study was to describe the experiences of pregnant women in prison and to assess the level of postpartum depression of women who give birth while incarcerated. All of the study participants had been screened for serious mental disorders upon arrival at VSPW. None of the women were deemed to have a serious mental disorder, including major depressive disorder, and thus were determined not to require services from a mental health professional before giving birth. By and large, this remained the case following pregnancy. None of the women were assessed to need psychotropic medication after giving birth. Although 17% of the 92 women completing the Beck Depression Inventory reported severe symptoms of depression, these symptoms appeared transitory and short term. With further clinical investigation, including mental status examination and diagnostic interviewing, these women were not shown to have significant impairments in mood and mood-disorder-related criteria (eating, sleeping, and energy) that would indicate a major depressive disorder with postpartum onset. It is noted that the Beck Depression Inventory was used as a subjective screening instrument along with other clinical data and not meant to be the decisive measurement of postpartum depression. Several studies note problems in using the Beck Depression Inventory to assess postpartum depression (Lee, 1997; Zerkowicz & Milet, 1995). A large number of somatic items assessed by the Beck Depression Inventory are items that might be commonly endorsed by postpartum women in general.

The women in this study had many factors that correlated with postpartum depression in other studies. More than 80% of the women were single, divorced, or separated. In 41% of the cases, the women reported that the father was either unaware of the pregnancy or would have no role in the child's life. The women were primarily unskilled laborers with low education levels. These women were expected to make life-changing decisions with limited resources and support.

In spite of numerous environmental factors that have been shown to lead to postpartum depression, the strongest correlations with postpartum depression were age of first pregnancy and drug of choice. The largest percentage of women reported that methamphetamine was their drug of choice. It is possible that there was a phenomenal underreporting of drug use given that the statistics were lower than any known documentation related to drug use and incarceration. For example, Fogel (1993) reported that 57.3% of pregnant inmates from a southern state prison reportedly smoked during their pregnancy, 27% used drugs, and 15.7% used alcohol. The drug pattern found in Fogel's study is current with that reported by other incarcerated women (Egley, et al., 1992; Fogel & Belyea, 1991).

In the present study, most participants said they received education about the potential effects of drug use during pregnancy. Looking at the global experience they reported, the researchers believe there was tremendous shame in reporting not only drug use during pregnancy but also any inadequacies as a mother. These women embrace the overriding societal message that no matter what the circumstances or resources, one must be a *good* mother.

Life Experiences of Pregnancy

The age range for first pregnancy was 14 to 31 years, with a mean of 18.5 years. The average number of pregnancies, 5.3, is high; however, this number makes sense taking into account the average numbers of living children and of deceased children. The average American family has

2.5 children; therefore, incarcerated women have more children than the average American family. Another difference is that although they have more children, not all of their children are in their custody. They reported very few visits with their newborn after giving birth because of traveling distance as well as concern for the babies' health and exposure to the prison.

Postbirth Contact With Newborns and Future Logistics

The location of the child after birth varied. The majority of babies were cared for by the inmate's immediate family. Others were living with the father or the inmate's extended family. A small number of babies were in foster care or Child Protective Services. Only 2% were adopted. Of this sample, very few pregnancies were aborted, ended in miscarriage, or were fetal deaths, stillborn births, or neonatal deaths. It is important to note the low numbers of abortion and adoption. Neither is used as an alternate means of dealing with a pregnancy and the stress of incarceration. Almost all babies were wanted. This finding is interesting considering the tremendous stress these mothers will face in prison and upon release and in light of the lack of resources and support that most of them reported.

A large majority of the mothers have full custody of their babies upon release. Very few were not allowed access to or information about their baby. Because of the numerous custody issues that some incarcerated women face, it would seem that fewer of these mothers would have custody of their newborn.

One third of the babies' fathers had a full-time role of caretaking, and 15% were caretaking part-time with family help. Almost half of the fathers were not involved in their babies' lives or did not know of the pregnancy. A small number of fathers were incarcerated at the time of the interview with the mothers. Three fourths of the participants anticipated financial support from a job, welfare, or a spouse or boyfriend. Only 7% did not know how they would be financially supported. Emotional support is mostly from family and friends, whereas 13% had no one for emotional support. These findings indicate that most babies were born into a single-parent family, exposing them to all the risks of this family type.

Future Plans for Pregnancy

A majority of the inmates said they did not want or were undecided about future children. When asked about pregnancy prevention, 62% said they planned to use birth control, with birth control pills and condoms being the most popular methods; 24% wanted a Tubule Ligation when released from prison (and some could have it done when giving birth in prison); and 8% had no plans for prevention. Therefore, almost all of the participants in this study were concerned about birth control and knew how to obtain it.

Thoughts About Pregnancy While Incarcerated

The majority of study participants reported that their overall experience of being pregnant in prison was negative. They missed their families and sharing the experience with significant others and family. They found the experience to be very stressful and did not want to be pregnant in prison again. However, other than stating very generally that they would not come to prison if pregnant, they had no other ideas about how to change their experience. Several also mentioned that they were unhappy about being separated from their baby. A small number stated that their experience was positive and that they were happy with the medical process.

When comparing data from this study with data collected in Mississippi in 1987 (LeFlore & Holston, 1990), it was found that some information was replicated and some

was dramatically different. The racial makeup is very different at VSPW compared to Mississippi. LeFlore and Holston (1990) reported 87% African American and 13% White. California's population is more diverse, which shows in the prison population also. Furthermore, marital status demographics are different across data from VSPW, Fogel (1993), and LeFlore and Holston (1990). California inmates have a higher rate of single marital status (73% vs. 48% and 44%, respectively) and a lower rate of separation or divorce, whereas marriage rates are about the same.

Another difference that stands out in comparison to these other studies is that VSPW has a significantly higher rate of children per incarcerated mother (5.3 children) than in Mississippi (2.7 children; LeFlore & Holston, 1990) or the Fogel (1993) study (3 children). This finding is surprising because more women at VSPW are single and yet are the same average age as the incarcerated women in the other studies. One possible explanation is the difference in average sentence length between VSPW, 2.3 years, and Mississippi, 7.5 years (LeFlore & Holston, 1990). It is also possible that these differences are a factor of cultural differences between California and Mississippi.

It is important to make a statement about the affective state of the participants in this study, especially with regard to the open-ended part of the interview. Almost all of the participants seemed disconnected from their feelings about the experience of motherhood in prison. It was not clear whether this was because they were in denial of the experience or situation or whether they were defending themselves with disassociation. They were very matter-of-fact when responding and did not show shame or remorse about being separated from their newborns. This could be due to the brief format of the interview or the lack of rapport in this short time period. They may be willing to discuss these feelings with persons with whom they feel close and safe to show their emotions. However, the general impression among those working in a women's prison is how incredibly open and unguarded women inmates usually are in terms of expressing their emotions and discussing their life experiences. This finding is interesting in light of the intimacy of the topic and the expectation or assumption that emotion would be expressed when discussing the separation of a mother from her newborn for an extended period of time.

Limitations of this study include the obvious problems with self-report measures, especially among incarcerated populations; the possible sampling bias created by the selection process (women who did not want to talk about their experience did not show for the interview); and the restricted use of the Beck Depression Inventory. Research has continued on this topic and will address these issues in more detail through methodological improvements based on this exploratory study. One area to be addressed is exploration of the lack of affective response and possible disassociation of the participants.

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